

Automated Survey Analytics & Reporting Platform

Data Product Optimization & End-to-End Workflow Automation [LIVE LINK](#) [GITHUB](#) [LINKEDIN](#)

PYTHON STREAMLIT PANDAS DATA ARCHITECTURE AUTOMATED BI

95% Faster

TIME EFFICIENCY

Turned days of manual spreadsheet slicing into under 5 seconds of automated execution.

0% Error Rate

DATA INTEGRITY

Eliminated human copy-paste errors via rigorous algorithmic data validation routines.

100k+ Rows

SCALABILITY

Engineered lightweight cross-filtering backend performing smoothly on enterprise-scale files.

Executive Summary & Business Context

Universities and organizations regularly capture massive volumes of evaluation data via web forms. However, translating raw response sheets into structured, shareable insights for stakeholders typically requires painful manual labor—building separate tables, calculating distributions, and drawing charts for dozens of distinct entities.

This project introduces a comprehensive, production-grade **Data Analytics Product** that completely automates the ingestion, structural classification, statistical processing, interactive visualization, and multi-format document distribution of survey datasets.

Technical Architecture & Core Components

1. Dynamic Ingestion & In-Memory Extraction

Processes raw, multi-format spreadsheets natively via Pandas. The backend analyzes dataframe metadata to intelligently classify column types into continuous rating scales (e.g., Likert scales 1-5) and categorical discrete variants, dynamically building downstream workflows.

2. Multi-Criteria Cross-Filtering Engine

Instead of static views, the interface exposes complex multidimensional slicing. Stakeholders can compound independent filtering vectors (e.g., isolating specific academic courses combined with unique instructors) with real-time reactive calculation.

Analyze responses

Set up filters

Filter 1

Q1. Select your course

Value 1

3rd Year (Bachelor)

☒ + Add second filter

Filter 2

Q2. Select professor

Value 2

Professor John Smith

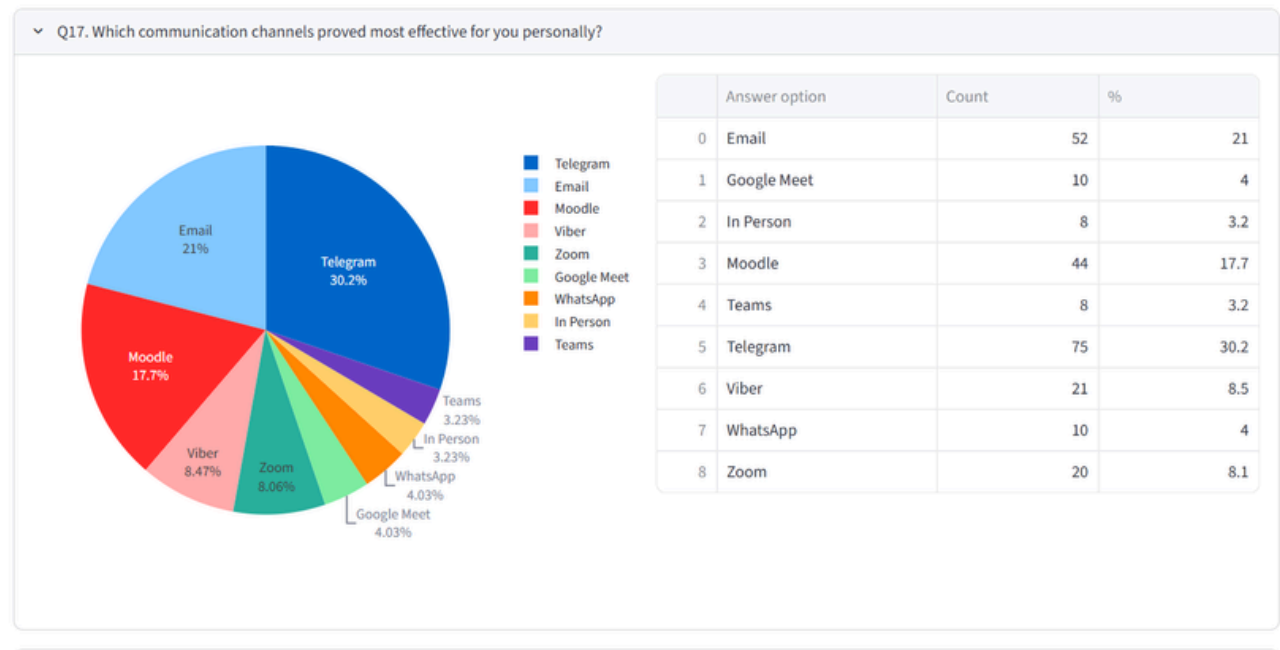
Question for analysis

Q4. How do you evaluate the professor's organizational culture (punctuality, time management)?

Apply filters

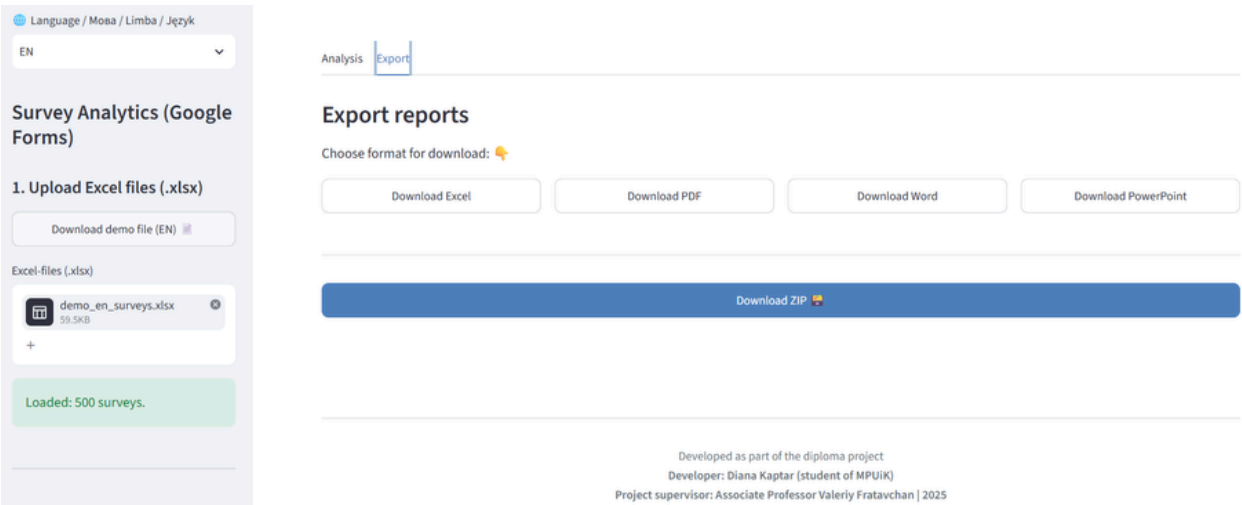
3. Automated Statistical Compiling

Computes complex frequency matrices, normalized distribution percentages, and cumulative statistical indices on-the-fly, serving as the trusted core compute engine for both visualizations and reporting modules.



4. Multi-Engine Document Compiler

Generates polished, executive-ready corporate files downloadable in a single click. Programmatically compiles structured outputs into **PDF**, **Microsoft Word (DOCX)**, **PowerPoint (PPTX)**, and **native-formatted Excel (XLSX)**.

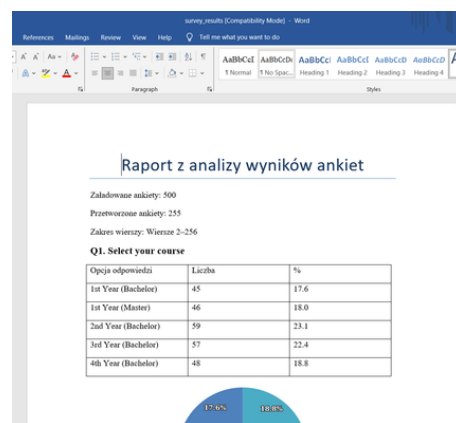
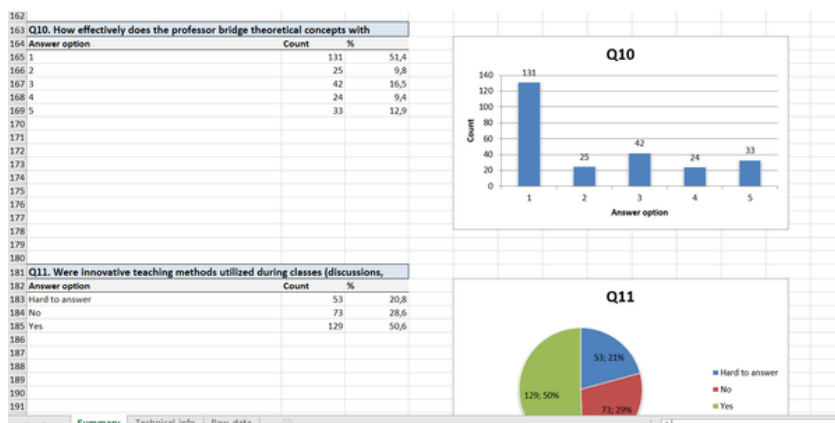


Performance Testing & Edge-Case Robustness

- **Volume Stress Testing:** Successfully benchmarked datasets scaling from 1,000 to 100,000+ distinct response nodes, maintaining snappy front-end interactivity and instant rendering under high load.
- **Algorithmic Guardrails:** Built custom exception handling layers targeting critical input boundaries (e.g., single-row micro-datasets, empty rows, invalid data types, and logical inversions where dynamic slice minima exceed maxima).

Business Value & Measurable Impact

- Optimized resource utilization by transitioning labor allocation away from low-value data preparation directly toward executive decision-making.
- Empowered non-technical university administrators to extract complex cross-filtered metrics independently, removing the historical dependency on dedicated database engineers.
- Delivered a robust, cross-platform interactive interface fully compatible with desktop and mobile viewport footprints.



Developer Profile & Stack Integration

Developer: Diana Kaptar — Web & Data Analyst

Links: [LIVE LINK](#), [GITHUB](#), [LINKEDIN](#)

Core Frameworks & Tools: Python 3.x | Streamlit Cloud Deployment | Pandas Engine | OpenPyXL Compiler | Git Version Control | Vector Visualization Engines